Pandas - Data Correlations

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https://github.com/Abdou-fi/python_libraries

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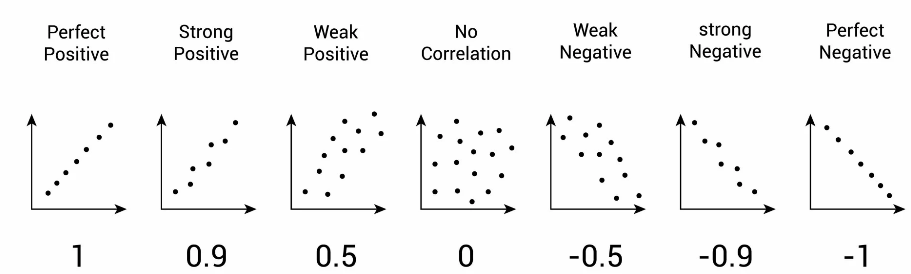
This is a Jupyter notebook converted to LaTeXusing the nbconvert tool. It contains explanation, code snippets and outputs related to data correlation using the ***Pandas*** library. For any question or remark please email me or send a request throught my github.

1 Finding Relationships



A great aspect of the Pandas module is the corr() method. The corr() method calculates the relationship between each column in your data set. The examples in this page uses a CSV file called: 'data.csv'.

[ ]:
```
import pandas as pd
df = pd.read_csv('data.csv')
#print(df.info())
df.corr()
```

[ ]:

	Duration	Pulse	Maxpulse	Calories
Duration	1.000000	-0.155408	0.009403	0.922717
Pulse	-0.155408	1.000000	0.786535	0.025121
Maxpulse	0.009403	0.786535	1.000000	0.203813
Calories	0.922717	0.025121	0.203813	1.000000

Note: The corr() method ignores "not numeric" columns.

2 Result Explained

The Result of the corr() method is a table with a lot of numbers that represents how well the relationship is between two columns. The number varies from -1 to 1. 1 means that there is a 1 to 1 relationship (a perfect correlation), and for this data set, each time a value went up in the first column, the other one went up as well. 0.9 is also a good relationship, and if you increase one value, the other will probably increase as well. -0.9 would be just as good relationship as 0.9, but if you increase one value, the other will probably go down. 0.2 means NOT a good relationship, meaning that if one value goes up does not mean that the other will.

What is a good correlation? It depends on the use, but I think it is safe to say you have to have at least 0.6 (or -0.6) to call it a good correlation.

3 Perfect Correlation:

We can see that “Duration” and “Duration” got the number 1.000000, which makes sense, each column always has a perfect relationship with itself.

4 Good Correlation:

“Duration” and “Calories” got a 0.922721 correlation, which is a very good correlation, and we can predict that the longer you work out, the more calories you burn, and the other way around: if you burned a lot of calories, you probably had a long work out.

5 Bad Correlation:

“Duration” and “Maxpulse” got a 0.009403 correlation, which is a very bad correlation, meaning that we can not predict the max pulse by just looking at the duration of the work out, and vice versa.